The Emergence of Co-Payment Burden Among Medical Scheme Members in South Africa

A Dual-Period Propensity Score Matching Analysis Using the IES 2010/11 and IES 2022/23

Statistical Analysis Report

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Abstract

Background: South Africa's two-tier healthcare system creates a structural divide between medical scheme members who access private care and the uninsured majority who rely on public facilities. Whether medical scheme membership generates additional out-of-pocket (OOP) co-payment burden beyond premiums — and whether this burden has changed over time — remains underexplored in the health financing literature.

Methods: Using microdata from two nationally representative Income and Expenditure Surveys conducted by Statistics South Africa — the IES 2010/11 (n = 21,802) and IES 2022/23 (n = 18,455) — we employ propensity score matching (PSM) via logistic regression with 1:1 nearest-neighbour matching on the logit propensity score to estimate the Average Treatment Effect on the Treated (ATT) of medical scheme membership on OOP health expenditure. Mahalanobis nearest-neighbour matching (NNM) is used as a sensitivity analysis. We additionally compute the revised SDG Indicator 3.8.2 (adopted March 2025), measuring the proportion of the population whose OOP health spending exceeds 40% of household discretionary budget, anchored to the Societal Poverty Line (SPL), testing sensitivity across four SPL definitions and disaggregating by scheme membership.

Results: Under PSM, the 2010/11 national ATT was statistically indistinguishable from zero (ATT = –R328; 95% CI: –R837 to R182; $p = 0.208$; $d = -0.025$), with matched controls exhibiting *higher* utilisation (95.1%) than treated units (85.3%). By 2022/23, a large and highly significant co-payment premium had emerged: ATT = R851 (95% CI: R721 to R982; $p < 10^{-37}$ $d = 0.159$), with treated utilisation (80.0%) now substantially exceeding controls (46.2%). NNM sensitivity analysis confirms both findings (2010/11: ATT = –R41, $p = 0.874$; 2022/23: ATT = R968, $p < 10^{-43}$). PSM achieves superior covariate balance (all post-matching SMDs < 0.02 in 2022/23) compared with NNM. The OOP distribution among uninsured households is genuinely bimodal: zero health expenditure rose from 11.7% to 58.9% between the two surveys.

The revised SDG 3.8.2 analysis reveals a striking reversal. In 2010/11, uninsured households experienced far higher financial hardship (37.4%) than insured households (1.5%) under the WHO SPL. By 2022/23, the relationship had inverted: insured households now show *higher* financial hardship rates (17.4%) than uninsured households (12.7%).

Conclusion: The co-payment burden associated with medical scheme membership has undergone a structural transformation, confirmed by both PSM and NNM. The 2010/11 null is robust to both estimators — the emergence of a significant ATT by 2022/23 is not a methodological artefact but a genuine structural shift. The revised SDG 3.8.2 indicator masks a protection paradox: the apparent decline is driven by suppressed demand among the uninsured, not genuine financial protection.

Keywords: out-of-pocket health expenditure, medical schemes, propensity score matching, nearest-neighbour matching, SDG 3.8.2, financial risk protection, South Africa, health financing, IES 2010/11, IES 2022/23

1 Introduction

South Africa's healthcare system is characterised by pronounced duality. Medical scheme members — comprising approximately 16–17% of the population — access a well-resourced private sector, while the remaining 83–84% rely primarily on the public health system (Council for Medical Schemes, 2023). This divide has deepened over recent decades despite policy interventions, including the Medical Schemes Act of 1998 and the ongoing National Health Insurance (NHI) Bill. Understanding how the financial burden of healthcare has evolved for both insured and uninsured populations is essential for informed policy design.

A central question in health financing concerns out-of-pocket (OOP) expenditure: the direct costs borne by households for healthcare beyond any insurance coverage. For medical scheme members, OOP spending includes co-payments, deductibles, above-threshold expenses, and costs for services not covered by their benefit packages. For non-members, OOP spending captures direct payments for consultations, medicines, and other healthcare services.

Naive comparisons of OOP expenditure between these groups are confounded by the strong socioeconomic gradient in medical scheme membership. Wealthier, more educated, and urban households are far more likely to belong to medical schemes, and these same characteristics predict higher healthcare utilisation and spending. Propensity score matching (PSM) addresses this confounding by estimating the probability of treatment and matching units with similar propensities (Rosenbaum & Rubin, 1983). PSM is particularly well-suited to this setting because the covariates are substantially correlated — income, expenditure, education, and population group are jointly determined by socioeconomic position — and PSM reduces these to a single balancing score. Mahalanobis nearest-neighbour matching (NNM), which matches directly on the multivariate covariate space, is used as a sensitivity analysis to confirm robustness.

While prior research has examined health expenditure inequality in South Africa (Ataguba & McIntyre, 2012; McIntyre *et al.*, 2009; Ataguba, 2012), Ataguba & Goudge (2012) employed propensity score methods to study the effect of health insurance on healthcare utilisation and OOP payments using the 2005/06 IES. Our use of PSM on the 2010/11 and 2022/23 waves provides a direct methodological extension, enabling temporal comparison with that earlier work.

This study exploits the availability of two nationally representative IES datasets — 2010/11 and 2022/23 — to examine five questions:

1.1 Objectives

1. Estimate the ATT of medical scheme membership on OOP health expenditure for both 2010/11 and 2022/23, nationally and by province.
2. Assess how the co-payment differential has changed over the 12-year period, adjusting for inflation.
3. Examine changes in healthcare utilisation patterns and covariate overlap between the two periods.
4. Characterise the distribution of OOP health expenditure among uninsured households and assess whether the mass at zero represents genuine non-utilisation.
5. Compute the revised SDG Indicator 3.8.2 (WHO, 2025b) for both periods, disaggregated by medical scheme membership, to assess the implications of the co-payment gap for financial risk protection and South Africa's progress toward SDG Target 3.8.

2 Data and Methods

2.1 Data Sources

2.1.1 IES 2010/11

The Income and Expenditure Survey 2010/11 was conducted by Statistics South Africa between September 2010 and August 2011. The survey employed a two-stage stratified sampling design with probability-proportional-to-size sampling of primary sampling units (PSUs) from the 2001 Census Master Sample. The sample comprised 31,419 dwelling units drawn from 3,254 PSUs. The national response rate was 91.6% (Stats SA, 2012). All expenditure values are annualised and deflated to March 2011 prices using the consumer price index.

2.1.2 IES 2022/23

The IES 2022/23 was conducted between November 2022 and November 2023 using a stratified multi-stage design based on the Geospatial Information Frame (GIF), with 31,042 dwelling units drawn from 3,318 PSUs. The national response rate was 81.94% (Stats SA, 2025). All values are annualised and adjusted to May 2023 prices using CPI deflators.

Both surveys use the COICOP (Classification of Individual Consumption According to Purpose) standard for expenditure classification and provide household-level survey weights calibrated to known population benchmarks.

2.2 Variables

Outcome variable: Total annual OOP health expenditure, computed as the sum of all household expenditure under COICOP Division 06 (Health), comprising medicines and health products (Group 061), outpatient care services (Group 062), inpatient care services (Group 063), and other health services (Group 064). Medical scheme premiums are classified separately under Division 12 (Insurance and Financial Services) and are excluded.

Treatment variable: Medical scheme membership, coded as a binary indicator. In the IES 2010/11, this is derived from the variable `q31021medaid` (medical aid membership, 1 = Yes, 2 = No). In the IES 2022/23, the variable is `eoh_meds` (1 = Yes, 2 = No). Households with missing, unspecified, or "not applicable" responses were excluded.

Matching covariates: The following eight covariates were used for matching in both periods, harmonised across the two survey instruments:

Covariate	Type	IES 2010/11	IES 2022/23
Log expenditure	Continuous	`consumptions`	`expenditure`
Log income	Continuous	`income`	`income`
Household size	Continuous	`hsize`	`hsize`
Head age	Continuous	Persons: `q14age`	`head_age`
Head sex	Binary	`genderofhead`	`head_sex`
Head population group	Categorical	`popgrpofhead`	`head_population`
Head education	Ordinal (5 groups)	Persons: `q21highestlevel`	`head_education`
Settlement type	Categorical	`settlement_type`	`settlement_type`

Table 1: Matching Covariates and Variable Harmonisation

Education levels (coded 0–27 in both surveys) were grouped into five ordinal categories: no schooling (0), primary (1–7), secondary (8–12), certificate/diploma (13–20), and degree/post-graduate (21–27). For the 2010/11 data, head demographics (age, education) were obtained by joining the household file with the persons file on the household head indicator (`q15relationship` = 1).

2.3 Propensity Score Matching (Primary Estimator)

The propensity score $e(\boldsymbol{x}) = \Pr(T = 1 \mid \boldsymbol{X} = \boldsymbol{x})$ is estimated via logistic regression of the treatment indicator on all eight covariates, fitted using iteratively reweighted least squares (IRLS). Covariates are standardised prior to estimation for numerical stability.

Matching is performed one-to-one without replacement on the logit of the estimated propensity score, $\text{logit}(\hat{e}(\boldsymbol{x}))$, using greedy nearest-neighbour matching with a caliper of 0.2 standard deviations of the logit propensity score distribution (Austin, 2011). Matching is restricted to the region of common support.

PSM is preferred over Mahalanobis distance matching for two reasons. First, the covariates exhibit substantial collinearity — income, expenditure, education, and population group are jointly determined by socioeconomic position — and PSM reduces these correlated covariates to a single balancing score, avoiding the curse of dimensionality. Second, PSM enables direct comparability with Ataguba & Goudge (2012), who used propensity score methods on the IES 2005/06, permitting a cleaner temporal comparison across three survey waves.

2.4 Mahalanobis NNM (Sensitivity Estimator)

As a sensitivity analysis, we also implement one-to-one nearest-neighbour matching without replacement using Euclidean distance on the standardised covariate vector:

$$j(i) = \operatorname*{argmin}_{j \in C} \left\| \frac{\boldsymbol{x}_i - \boldsymbol{x}_j}{\boldsymbol{s}} \right\|$$

where $\boldsymbol{s}$ is the pooled standard deviation vector and C is the set of unmatched control units. A caliper of 0.5 pooled standard deviations per dimension was applied at the national level, relaxed to 0.75 for provincial analyses. Matching was performed greedily in random order (seed = 42) without replacement. NNM provides a stricter matching criterion than PSM and is used to verify that the PSM findings are not driven by covariate-specific imbalances masked by the propensity score.

2.5 Estimand and Inference

The estimand is the Average Treatment Effect on the Treated (ATT):

$$\text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid T = 1]$$

estimated as:

$$\widehat{\text{ATT}} = \frac{1}{N_m} \sum_{i \in \text{ matched}} \left[Y_i^T - Y_{j(i)}^C \right]$$

Standard errors are computed as $\sigma_d / \sqrt{N_m}$, where σ_d is the standard deviation of matched pair differences. We report paired t-tests, Wilcoxon signed-rank tests, 95% confidence intervals (analytical and bootstrap with 500 replications), and Cohen's d as a standardised effect size measure.

2.6 Balance Diagnostics

Covariate balance is assessed using standardised mean differences (SMD). An absolute SMD below 0.10 is conventionally considered adequate (Austin, 2011). We report SMDs before and after matching, together with percentage reduction.

2.7 Inflation Adjustment

For cross-period comparisons, all values are expressed in May 2023 prices. The CPI adjustment factor from March 2011 to May 2023 is 1.727, derived from the Stats SA headline CPI (all items, metropolitan and other urban areas, December 2016 = 100 base): index values of 67.8 (March 2011) and 117.1 (May 2023).

2.8 Sensitivity Analyses

For each survey period, four sensitivity analyses are conducted:

1. **Mahalanobis NNM:** multivariate matching on the full covariate vector, as described above.
2. **Bootstrap confidence intervals** (500 replications) for the PSM national ATT.
3. **Exact province matching:** PSM is performed separately within each province and results are pooled with sample-size weighting.
4. **Positive health expenditure only:** restricting the sample to households with OOP > 0 to assess the effect among healthcare utilisers.

2.9 Samples

After exclusions for missing treatment status, zero expenditure, invalid demographic data, and unspecified population group or education, the analysis samples are:

Characteristic	IES 2010/11	IES 2022/23
Total households	21,802	18,455
Medical scheme members (treated)	4,096	7,711
Non-members (control)	17,706	10,744
Sample % treated	18.8%	41.8%
Weighted % treated	21.8%	42.5%
Mean annual expenditure (R)	83,112	121,334
Mean annual income (R)	104,296	166,665
Survey response rate	91.6%	81.9%

Table 2: Analysis Samples

3 Results

3.1 IES 2010/11: National Analysis

3.1.1 Propensity Score Estimation

The logistic regression model for propensity score estimation achieves good discrimination between treated and control units: the mean propensity score is 0.541 for treated and 0.106 for control households, reflecting the substantial socioeconomic distance between scheme members and non-members in 2010/11. The caliper of $0.2 \times \text{SD}(\text{logit PS}) = 0.427$ on the logit scale accommodates this wide separation while enforcing closeness within matched pairs. The common support region spans [0.001, 0.997], and only 2 treated units fall outside it.

3.1.2 Matching Performance

Of 4,096 treated households, 2,586 (63.1%) were successfully matched under PSM. The match rate is somewhat lower than the NNM rate (66.0%) because the single-dimensional propensity score caliper imposes a different constraint than the per-dimension Mahalanobis caliper.

3.1.3 Covariate Balance

Covariate	**SMD Before**	**SMD After**	**% Reduction**
Log expenditure	1.8304	–0.0059	99.7%
Log income	1.3221	0.0082	99.4%
Household size	–0.2408	0.0141	94.1%
Head age	–0.0567	0.0179	68.5%
Head sex	–0.3197	–0.0129	96.0%
Head population group	0.8703	0.0020	99.8%
Education group	0.6547	0.0592	91.0%
Settlement type	–0.7176	0.0253	96.5%

Table 3: Covariate Balance — IES 2010/11 National (PSM)

PSM achieves substantially better covariate balance than NNM in 2010/11. The most dramatic improvement is in log expenditure: the post-matching SMD is –0.006 under PSM (versus 0.179 under NNM), a reduction from 90.2% to 99.7%. This is a key advantage of PSM over NNM — the propensity score integrates all covariates into a single balancing score, achieving near-perfect balance on the most important confounder (expenditure) even when common support is limited. The education group SMD (0.059) exceeds the 0.05 threshold but remains well within acceptable bounds.

3.1.4 Average Treatment Effect on the Treated

Estimate	**Value**
Matched pairs	2,586
Mean OOP — treated (R)	2,204.34
Mean OOP — matched control (R)	2,531.85
ATT (R)	**–327.51**
Standard error (R)	260.09
95% CI — analytical (R)	[–837.29, 182.28]
95% CI — bootstrap (R)	[–818.31, 185.16]
Paired t-statistic	–1.259
p-value (paired t-test)	0.208
Cohen's d	–0.025

Table 4: National ATT Estimates — IES 2010/11 (PSM)

The national ATT in 2010/11 is statistically indistinguishable from zero under PSM (ATT = –R328; $p = 0.208$; $d = –0.025$). The point estimate is negative and somewhat larger in magnitude than the NNM estimate (–R41), but the confidence interval comfortably spans zero. The NNM sensitivity analysis confirms the null finding (ATT = –R41; $p = 0.874$). The naive (unmatched) difference of R2,106 dramatically overstates the matched estimate, confirming the severity of selection bias.

The PSM point estimate suggests that, if anything, comparable non-members may have incurred *higher* OOP expenditure than scheme members in 2010/11. This is consistent with the utilisation pattern described below and with the high propensity of non-members to use public-sector healthcare, where co-payments for services such as over-the-counter medicines and traditional remedies contributed to OOP spending.

3.1.5 Healthcare Utilisation

A notable finding is the high utilisation rate among matched controls: 95.1% of matched non-member households reported some health expenditure, compared with 85.3% of matched scheme member households. This pattern — higher utilisation among controls — is the opposite of what is observed in the 2022/23 data and suggests that in 2010/11, comparable non-member households accessed healthcare at similar or higher rates.

3.2 IES 2022/23: National Analysis

3.2.1 Propensity Score Estimation

The propensity score model for 2022/23 shows substantially better overlap than 2010/11: the mean PS is 0.470 for treated and 0.380 for control units. The caliper = 0.133 on the logit scale (0.2 × SD). Only 3 treated units fall outside the common support region [0.084, 0.892].

3.2.2 Matching Performance

Of 7,711 treated households, 6,487 (84.1%) were successfully matched under PSM, comparable to the NNM rate (85.9%). The improved overlap relative to 2010/11 reflects the narrower socioeconomic gap between scheme members and non-members in the more recent period.

3.2.3 Covariate Balance

Covariate	SMD Before	SMD After	% Reduction
Log expenditure	0.6132	0.0020	99.7%
Log income	0.4571	–0.0063	98.6%
Household size	0.1766	–0.0188	89.4%
Head age	0.1641	0.0040	97.6%
Head sex	0.0629	–0.0006	99.0%
Head population group	0.2103	0.0132	93.7%
Education group	0.2369	0.0071	97.0%
Settlement type	–0.0593	0.0039	93.4%

Table 5: Covariate Balance — IES 2022/23 National (PSM)

PSM achieves excellent balance: all post-matching SMDs are below 0.02 (mean |SMD| = 0.007), substantially better than NNM (mean |SMD| = 0.021). The largest residual imbalance is household size (–0.019), well within the 0.05 threshold for negligible bias. The propensity score collapses the eight correlated covariates into a single balancing score, achieving near-perfect balance simultaneously across all dimensions.

3.2.4 Average Treatment Effect on the Treated

Estimate	**Value**
Matched pairs	6,487
Mean OOP — treated (R)	1,693.30
Mean OOP — matched control (R)	841.89
ATT (R)	**851.41**
Standard error (R)	66.57
95% CI — analytical (R)	[720.93, 981.89]
95% CI — bootstrap (R)	[712.68, 991.04]
Paired t-statistic	12.789
p-value (paired t-test)	5.23×10^{-37}
Cohen's d	0.159

Table 6: National ATT Estimates — IES 2022/23 (PSM)

Under PSM, the 2022/23 ATT is R851 ($p < 10^{-37}$ $d = 0.159$) — large, highly significant, and robust. The NNM sensitivity analysis yields a somewhat larger estimate (R968, $d = 0.170$), which is expected: NNM matches more tightly on expenditure, which creates slightly different counterfactuals. The PSM and NNM confidence intervals overlap substantially, confirming the robustness of the finding.

The PSM estimate (R851) is more conservative than NNM (R968) because PSM achieves better balance on expenditure (SMD 0.002 vs 0.090), reducing residual confounding from income differences between matched pairs.

3.2.5 Healthcare Utilisation

The utilisation pattern has reversed: 80.0% of matched treated households report any health spending versus only 46.2% of matched controls. The median OOP expenditure is R457 for treated households versus R0 for controls — a stark indicator of suppressed demand among non-members.

3.3 Provincial Analysis

3.3.1 IES 2010/11

Province	**Pairs**	**ATT (R)**	**SE (R)**	**95% CI (R)**	***p*-value**	**Cohen's *d***
Western Cape	477	–1,776.76	1,206.15	[–4,141, 587]	0.141	–0.067
Eastern Cape	257	–304.67	235.81	[–767, 158]	0.198	–0.081
Northern Cape	128	–843.75	365.06	[–1,559, –128]	2.2×10^{-2}	–0.204
Free State	202	–492.24	473.51	[–1,420, 436]	0.300	–0.073
KwaZulu-Natal	298	–412.61	326.18	[–1,052, 227]	0.207	–0.073
North West	256	623.77	409.18	[–178, 1,426]	0.129	0.095
Gauteng	456	768.81	418.39	[–51, 1,589]	0.067	0.086
Mpumalanga	201	144.38	313.25	[–470, 759]	0.645	0.033
Limpopo	215	174.08	230.22	[–277, 625]	0.450	0.052

Table 7: Provincial ATT Estimates — IES 2010/11 (PSM)

Under PSM, **no province** shows a statistically significant positive ATT in 2010/11 — not even Gauteng ($p = 0.067$) or Mpumalanga ($p = 0.645$), which showed significance under NNM. The Northern Cape shows a significant *negative* ATT (–R844; $p = 0.022$), suggesting that scheme members in this province incurred *less* OOP expenditure than comparable non-members. This is consistent with effective insurance protection in a province with limited private healthcare infrastructure.

The PSM results for 2010/11 are more uniformly null than the NNM results. Under NNM, two provinces (Gauteng, Mpumalanga) showed significance, but the PSM analysis — with its superior balance — eliminates these. This suggests that the NNM significance in those provinces may have been driven by residual confounding from imperfect balance on expenditure (SMD = 0.179 under NNM vs –0.006 under PSM).

3.3.2 IES 2022/23

Province	**Pairs**	**ATT (R)**	**SE (R)**	**95% CI (R)**	***p*-value**	**Cohen's *d***
Western Cape	604	1,412.18	329.95	[765, 2,059]	2.2×10^{-5}	0.174
Eastern Cape	871	719.11	100.63	[522, 916]	1.9×10^{-12}	0.242
Northern Cape	247	791.98	327.96	[149, 1,435]	1.7×10^{-2}	0.154
Free State	488	345.27	144.47	[62, 628]	1.7×10^{-2}	0.108
KwaZulu-Natal	1,068	968.67	143.13	[688, 1,249]	2.2×10^{-11}	0.207
North West	383	503.51	129.71	[249, 758]	1.2×10^{-4}	0.198
Gauteng	1,531	459.45	158.90	[148, 771]	3.9×10^{-3}	0.074
Mpumalanga	511	658.35	127.34	[409, 908]	3.4×10^{-7}	0.229
Limpopo	656	761.48	123.38	[520, 1,003]	1.2×10^{-9}	0.241

Table 8: Provincial ATT Estimates — IES 2022/23 (PSM)

Under PSM, the ATT is statistically significant ($p < 0.05$) in **all nine provinces** in 2022/23, confirming the NNM finding. The PSM estimates are generally more conservative (mean provincial ATT: R736 under PSM vs R917 under NNM), reflecting better covariate balance. The largest effects are in the Western Cape (R1,412; $d = 0.174$), KwaZulu-Natal (R969; $d = 0.207$), and Northern Cape (R792; $d = 0.154$). The smallest effect is in Free State (R345; $d = 0.108$) and Gauteng (R459; $d = 0.074$).

3.4 Cross-Period Comparison

3.4.1 National ATT in Real Terms

Measure	IES 2010/11*	IES 2022/23	Change
ATT — PSM (R)	–565.72	851.41	+1,417
ATT — NNM (R)	–70.59	968.44	+1,039
Naive difference (R)	3,637.43	1,679.68	–1,958
Matched treated mean (R)	3,807.12	1,693.30	–55.5%
Matched control mean (R)	4,372.53	841.89	–80.7%
% any health spend — treated	85.3%	80.0%	–5.3pp
% any health spend — control	95.1%	46.2%	–48.9pp

Table 9: Cross-Period Comparison of National ATT (May 2023 Prices)

* 2010/11 values inflated to May 2023 prices using CPI factor of 1.727 (Stats SA P0141). Original values are in March 2011 prices.

The cross-period comparison reveals three striking findings:

First, the ATT shifted from zero to R851 under PSM (R968 under NNM). In 2010/11, both estimators find no significant co-payment differential. By 2022/23, both find a large, highly significant effect. The consistency across methods confirms that the co-payment burden is a genuine structural feature, not a methodological artefact.

Second, the matched control mean declined by 81% in real terms (from R4,373 to R842 under PSM). Matched treated spending declined by 56% (from R3,807 to R1,693). The emergence of the ATT is driven entirely by the steeper decline among non-members.

Third, healthcare utilisation among matched controls collapsed. In 2010/11, 95.1% of PSM-matched controls reported some health expenditure. By 2022/23, this had fallen to 46.2% — a decline of 49 percentage points. Among treated households, the decline was modest (85.3% to 80.0%).

3.4.2 Provincial Evolution

Province	PSM 2011* (R)	PSM 2023 (R)	Change (R)	*d* 2011	*d* 2023
Western Cape	–3,068.87	1,412.18	+4,481	–0.067	0.174
Eastern Cape	–526.17	719.11	+1,245	–0.081	0.242
Northern Cape	–1,457.48	791.98	+2,249	–0.204	0.154
Free State	–850.31	345.27	+1,196	–0.073	0.108
KwaZulu-Natal	–712.60	968.67	+1,681	–0.073	0.207
North West	1,077.24	503.51	–574	0.095	0.198
Gauteng	1,327.77	459.45	–868	0.086	0.074
Mpumalanga	249.33	658.35	+409	0.033	0.229
Limpopo	300.62	761.48	+461	0.052	0.241

Table 10: Provincial ATT Evolution (May 2023 Prices)

* 2010/11 PSM ATT values inflated to May 2023 prices. Note: no province shows a significant positive PSM ATT in 2010/11.

Under PSM, the provincial pattern is starker than under NNM. All provinces except North West and Gauteng shifted from negative to positive ATTs — the co-payment premium is a new phenomenon across the country. The two provinces that retain positive (though non-significant) ATTs in 2010/11 (North West, Gauteng) show declining ATTs by 2022/23, suggesting convergence as scheme membership broadened.

3.5 Sensitivity Analyses

Analysis	ATT (R)	95% CI (R)	ATT (R)	95% CI (R)
	IES 2010/11		IES 2022/23	
PSM (primary)	**–327.51**	[–837, 182]	**851.41**	[721, 982]
PSM bootstrap (500 reps)	–240.15	[–818, 185]	846.78	[713, 991]
PSM exact province (pooled)	–271.63	[–775, 232]	740.92	[616, 866]
PSM positive OOP only	–321.48	$p = 0.242$	612.83	$p < 0.001$
NNM (sensitivity)	**–40.87**	[–545, 463]	**968.44**	[831, 1,106]
NNM bootstrap (500 reps)	46.93	[–528, 487]	951.23	[815, 1,093]

Table 11: Sensitivity Analysis Results — Both Periods

For the IES 2010/11, all analyses — PSM and NNM — confirm the null finding. No specification yields a significant positive ATT. The PSM estimates are consistently negative (–R241 to –R328) while the NNM estimates are close to zero (–R41 to +R47), but all confidence intervals comfortably span zero. The consistency across two fundamentally different matching approaches — one collapsing covariates to a single score, the other matching on the full multivariate space — strengthens confidence that the 2010/11 null is genuine.

For the IES 2022/23, all analyses confirm a significant positive ATT. The PSM range (R613 to R851) is somewhat more conservative than the NNM range (R951 to R968), reflecting PSM's superior balance on the key expenditure confounder. Even among healthcare utilisers only (PSM: R613; $p < 0.001$), scheme members pay substantially more than comparable non-members.

3.6 Covariate Overlap and Common Support

The declining pre-matching SMDs between the two periods are themselves an important finding. Table 12 summarises the maximum covariate imbalances.

Covariate	**SMD 2010/11**	**SMD 2022/23**
Log expenditure	1.830	0.613
Log income	1.322	0.457
Head population group	0.870	0.210
Settlement type	–0.718	–0.059
Education group	0.655	0.237
Head sex	–0.320	0.063
Household size	–0.241	0.177
Head age	–0.057	0.164
Mean |SMD|	0.752	0.247
PSM match rate	63.1%	84.1%
NNM match rate	66.0%	85.9%

Table 12: Pre-Matching Covariate Imbalance: 2010/11 vs 2022/23

The mean absolute pre-matching SMD declined from 0.752 to 0.247 — a 67% reduction. This indicates that the socioeconomic profile of medical scheme members converged substantially toward that of non-members between 2010/11 and 2022/23. The most dramatic improvement was in log expenditure (1.83 to 0.61), population group (0.87 to 0.21), and settlement type (0.72 to 0.06). This convergence enabled better matching (66% vs 86% match rate) and more credible causal estimates in the later period.

3.7 Distribution of OOP Health Expenditure Among Uninsured Households

A critical question for interpreting the ATT estimates — and for the SDG 3.8.2 analysis that follows — is whether the OOP distribution among uninsured households is truly bimodal (a genuine mass at zero and a separate positive-spending subpopulation) or whether it reflects a continuous distribution with many small values rounded or truncated to zero.

3.7.1 Distributional Characterisation

Measure	IES 2010/11	(weighted)	IES 2022/23	(weighted)
Total uninsured households	17,976		11,345	
Zero OOP (count)	2,100		6,683	
Zero OOP (%)	11.7%	10.6%	58.9%	58.2%
Positive OOP (%)	88.3%	89.4%	41.1%	41.8%
Mean OOP — all (R)	815.23		517.89	
Mean OOP — positive only (R)	923.06		1,260.29	
Median OOP — all (R)	299.00		0.00	
Median OOP — positive only (R)	359.00		452.06	
Minimum positive value (R)	2.00		2.01	
Skewness (positive)	10.16		15.81	
Kurtosis (positive)	172.73		401.91	

Table 13: OOP Distribution Among Uninsured Households

3.7.2 Bimodality Assessment

The distribution is **genuinely bimodal**, not an artefact of rounding or truncation. Three lines of evidence support this:

First, the gap between zero and the smallest positive values is clean. In both surveys, the minimum positive OOP is approximately R2 per annum — not a fraction of a cent that might suggest rounding. There are no values between R0 and R2 in either period. This discrete gap confirms that zero represents true non-utilisation: these households have *no* COICOP Division 06 expenditure line items whatsoever.

Second, the mass at zero is large and has grown dramatically. The proportion of uninsured households with zero OOP increased from 11.7% (weighted: 10.6%) in 2010/11 to 58.9% (weighted: 58.2%) in 2022/23. This 47-percentage-point shift is far too large to be attributable to data processing changes and represents a genuine behavioural shift in healthcare utilisation.

Third, the positive-spending subpopulation has a well-defined distribution. Among the 41.1% of uninsured households with positive OOP in 2022/23, the distribution is heavily right-skewed (skewness = 15.81, kurtosis = 401.91) with a median of R452 and a long right tail extending to R123,362. This shape is characteristic of health expenditure distributions globally and shows no evidence of truncation or artificial compression at the lower end.

Expenditure Range	IES 2010/11	IES 2022/23
R0 (zero)	11.7%	58.9%
R0.01–R10	0.5%	0.5%
R10–R50	4.8%	2.6%
R50–R100	6.9%	2.4%
R100–R500	40.8%	15.7%
R500–R1,000	17.1%	7.9%
R1,000–R5,000	16.1%	9.3%
> R5,000	2.0%	2.0%

Table 14: Distribution of Positive OOP Expenditure Among Uninsured Households

3.7.3 Implications for Analysis

The bimodal distribution has two important analytical implications. First, it validates the NNM approach, which compares mean OOP levels: the ATT captures both the intensive margin (how much utilisers spend) and the extensive margin (the probability of any spending). The 2022/23 ATT of R968 is substantially driven by the extensive margin — the 34-percentage-point gap in utilisation rates between matched treated (80.2%) and matched control (46.6%) households.

Second, the bimodality creates a fundamental interpretive challenge for the SDG 3.8.2 indicator. As the conceptual framework identifies (Feedback 2: Poverty ↔ Ill Health), zero OOP does not imply financial protection — it may indicate suppressed demand due to access barriers. The 58.2% of uninsured households with zero health spending are not "protected"; they are not accessing care. This distinction is central to the analysis that follows.

3.8 Financial Risk Protection: SDG Indicator 3.8.2

3.8.1 Methodology

We compute SDG Indicator 3.8.2 using the revised methodology adopted by the UN Statistical Commission in March 2025 (WHO, 2025b). The revised indicator measures the proportion of the population with positive out-of-pocket household expenditure on health exceeding 40% of the household discretionary budget.

The discretionary household budget is defined as total household consumption expenditure per capita minus the societal poverty line (SPL), measured on a per capita daily basis:

$$\text{Indicator} = \frac{\sum_i m_i \omega_i \mathbf{1}\big(oop_i^{\text{health}} > 0.4 \cdot (y_i - \text{SPL}) \cap oop_i^{\text{health}} > 0\big)}{\sum_i m_i \omega_i}$$

where m_i is household size, ω_i is the sampling weight, oop_i^{health} is daily per capita OOP health expenditure, and y_i is daily per capita total consumption expenditure. For households living below the SPL ($y_i < \text{SPL}$), the discretionary budget is negative and any positive OOP expenditure constitutes financial hardship.

Using 2017 purchasing power parities, the WHO SPL is defined as:

$$\text{SPL} = \max\big(\$2.15/\ \text{day}, \$1.15/\ \text{day} + 0.5 \times \text{median}^*\big(y_i - oop_i^{\text{health}}\big)\big)$$

where the median is population-weighted and computed from per capita daily consumption excluding OOP health expenditure (WHO, 2025b; Jolliffe & Prydz, 2021).

To test sensitivity to the poverty line assumption, we compute the indicator under four SPL definitions:

SPL Variant	Definition	IES 2010/11	IES 2022/23
WHO SPL	max(IPL; relative formula)	R532/month	R1,117/month
Stats SA FPL	Food Poverty Line	R335/month	R760/month
Stats SA LBPL	Lower-Bound Poverty Line	R501/month	R1,058/month
Stats SA UBPL	Upper-Bound Poverty Line	R779/month	R1,558/month

Table 15: SPL Variants Used for SDG 3.8.2 Computation

We also report the original budget-share indicators (OOP > 10% and > 25% of total consumption) for comparison with the pre-revision methodology. All amounts are measured per capita per day, with population weighting ($m_i \times \omega_i$), as specified in the WHO metadata.

3.8.2 National Results: Original Budget-Share Approach

Under the original SDG 3.8.2 methodology, financial hardship rates are strikingly low:

	IES 2010/11			IES 2022/23		
Threshold	**All**	**Insured**	**Uninsured**	**All**	**Insured**	**Uninsured**
> 10% of consumption	1.7%	2.1%	1.6%	0.5%	0.7%	0.3%
> 25% of consumption	0.1%	0.4%	0.1%	0.0%	0.1%	0.0%

Table 16: Original SDG 3.8.2 — Budget-Share Approach

The original indicator paints a picture of near-universal financial protection: fewer than 2% of the population exceed the 10% threshold in either period. This is precisely the measurement artefact that motivated the 2025 revision. The budget-share approach is insensitive to poverty — a household spending R50 out of R5,000 (1%) appears equally protected as a household spending R5 out of R50 (10%), despite the latter being destitute. The decline from 1.7% to 0.5% between the two surveys reflects the suppression of healthcare demand among the poor, not improved protection.

3.8.3 National Results: Revised SDG 3.8.2 (40% of Discretionary Budget)

The revised indicator reveals a fundamentally different — and more concerning — picture:

	IES 2010/11			IES 2022/23		
SPL Variant	**All**	**Insured**	**Uninsured**	**All**	**Insured**	**Uninsured**
Stats SA FPL	15.8%	0.4%	19.3%	7.5%	8.6%	6.6%
Stats SA LBPL	28.8%	1.3%	35.2%	13.5%	15.7%	11.7%
WHO SPL	30.6%	1.5%	37.4%	14.8%	17.4%	12.7%
Stats SA UBPL	43.6%	4.0%	52.7%	23.5%	28.3%	19.5%

Table 17: Revised SDG 3.8.2 — 40% of Discretionary Budget by SPL Variant

Three findings stand out.

First, the revised indicator yields dramatically higher rates than the original. Under the WHO SPL, 14.8% of the population in 2022/23 experience financial hardship from health spending — 30 times higher than the original 10%-threshold rate (0.5%). This confirms the critique that the budget-share approach systematically understates the burden of health spending on the poor (Wagstaff, 2019; Grépin *et al.*, 2020).

Second, a remarkable reversal has occurred between insured and uninsured households. In 2010/11, the pattern was intuitive: uninsured households experienced far higher financial hardship than insured households across all SPL variants (e.g., 37.4% vs 1.5% under WHO SPL). By 2022/23, this relationship had **inverted**: insured households now show *higher* financial hardship rates (17.4%) than uninsured households (12.7%) under the WHO SPL. This inversion is consistent across all SPL variants.

Third, the overall rate has fallen, but this decline is misleading. The national rate dropped from 30.6% to 14.8% (WHO SPL). However, this apparent improvement is driven almost entirely by the collapse in healthcare utilisation among uninsured households documented in Table 13. A household with zero OOP cannot exceed any threshold — it is classified as "protected" by the indicator regardless of whether it has actually accessed needed care. The 58% of uninsured households with zero health spending in 2022/23 are counted as experiencing no financial hardship, even though their zero spending may reflect foregone care rather than genuine protection.

3.8.4 The Insured–Uninsured Reversal

The reversal in SDG 3.8.2 rates between insured and uninsured households is the central finding of this analysis and warrants detailed examination.

In 2010/11, insured households had near-zero financial hardship rates (0.4–4.0% depending on SPL) because they were overwhelmingly above all poverty lines: only 1.3% of insured households fell below the WHO SPL, and their OOP spending, while positive, rarely exceeded 40% of their ample discretionary budgets.

By 2022/23, two forces produced the reversal:

1. **Broadening of the insured base.** The expansion of medical scheme membership to lower-income households meant that 22.0% of insured households now fall below the WHO SPL (up from 1.3%). These newly insured, near-poor households face co-payments that can exceed 40% of their limited discretionary budgets.

2. **Demand suppression among the uninsured.** The collapse of healthcare utilisation from 88.3% to 41.1% among uninsured households mechanically reduced their SDG 3.8.2 rates: households that do not spend on healthcare cannot experience financial hardship as measured by the indicator.

This creates what might be termed the **protection paradox**: the indicator registers improved "protection" for uninsured households precisely when they are accessing *less* care. As the literature review identifies, zero OOP does not equal protection when it reflects suppressed demand (O'Donnell, 2024; Koch & Setshegetso, 2020; Ataguba & Goudge, 2012).

3.8.5 Sensitivity to SPL Choice

The absolute level of the revised indicator is highly sensitive to the poverty line assumed, but the **qualitative findings are robust** across all four SPL variants:

1. The insured–uninsured reversal appears under all four SPL definitions in 2022/23.
2. The decline from 2010/11 to 2022/23 appears under all four SPL definitions.
3. The ranking of provinces is consistent regardless of SPL choice.

The sensitivity is driven primarily by the share of households classified as below the SPL: under the FPL (the most conservative line), 15.4% of all households fall below the line; under the UBPL, this rises to 44.1%. Since any below-SPL household with positive OOP automatically qualifies as experiencing financial hardship, the choice of poverty line has a multiplicative effect on the indicator.

3.8.6 Provincial Analysis (Revised SDG 3.8.2, WHO SPL)

	IES 2010/11			IES 2022/23		
Province	**All**	**Insured**	**Uninsured**	**All**	**Insured**	**Uninsured**
Western Cape	13.8%	0.8%	18.4%	5.9%	6.0%	5.7%
Eastern Cape	36.7%	1.1%	43.0%	18.9%	23.6%	14.9%
Northern Cape	32.6%	5.7%	42.2%	13.8%	20.9%	10.2%
Free State	32.5%	2.6%	39.3%	17.0%	18.5%	15.5%
KwaZulu-Natal	44.4%	2.1%	51.0%	21.7%	25.1%	18.7%
North West	30.6%	2.1%	35.9%	21.6%	26.0%	19.2%
Gauteng	15.0%	0.7%	20.8%	8.4%	10.6%	6.4%
Mpumalanga	39.2%	1.7%	45.3%	17.7%	21.9%	13.9%
Limpopo	38.7%	4.2%	41.9%	16.5%	18.9%	14.8%

Table 18: Revised SDG 3.8.2 by Province (WHO SPL, 40% Threshold)

The provincial results reveal:

- **The insured–uninsured reversal is present in all nine provinces in 2022/23.** In every province, insured households now have higher SDG 3.8.2 rates than uninsured households — a pattern that was absent in 2010/11 when the opposite held universally.
- **The provinces with the largest NNM-estimated ATTs also show the largest insured SDG 3.8.2 rates.** The Eastern Cape (ATT = R1,046; insured SDG 3.8.2 = 23.6%), North West (ATT = R402; 26.0%), and KwaZulu-Natal (ATT = R1,133; 25.1%) combine high co-payment burdens with high financial hardship, indicating that the causal co-payment effect identified by NNM translates directly into measured financial risk.
- **The decline in national rates masks divergent trends.** The Western Cape saw the largest reduction in financial hardship (from 13.8% to 5.9%), consistent with its relatively high income levels. KwaZulu-Natal, despite a decline from 44.4% to 21.7%, retains the highest overall rate — consistent with its having both high poverty rates and significant co-payment exposure.

4 Discussion

4.1 Principal Findings

This study documents a structural transformation in the relationship between medical scheme membership and out-of-pocket health expenditure in South Africa over a 12-year period. Three principal findings emerge.

First, the co-payment burden associated with medical scheme membership has emerged as a new phenomenon. In 2010/11, both PSM (ATT = –R328, $p = 0.208$) and NNM (ATT = –R41, $p = 0.874$) find no significant effect. By 2022/23, PSM yields an ATT of R851 ($p < 10^{-37}$) and NNM yields R968 ($p < 10^{-43}$), both significant in all nine provinces. The consistency across two fundamentally different matching estimators — one that collapses covariates to a single propensity score, another that matches on the full multivariate space — strengthens the causal interpretation. The 2010/11 null is not a methodological artefact of the NNM approach: PSM, with its superior balance on expenditure (SMD = –0.006 vs 0.179), confirms the null.

Second, the emergence of the ATT is driven primarily by a collapse in healthcare utilisation among non-members. PSM-matched control utilisation fell from 95.1% to 46.2% between the two surveys — a decline of 49 percentage points. Matched treated utilisation declined only modestly (85.3% to 80.0%). The matched control mean OOP fell by 81% in real terms, compared with 56% for matched treated households. The ATT in 2022/23 thus reflects not an increase in scheme member spending but a steeper decline in non-member spending.

Third, the socioeconomic gradient in scheme membership has narrowed substantially. The mean pre-matching SMD across covariates fell from 0.75 to 0.25, and the match rate improved from 66% to 86%. This convergence likely reflects expansion of scheme membership to middle-income households, particularly through low-cost benefit options and employer-mandated coverage, as well as the changing employment and education profile of South African households.

4.2 Interpretation

4.2.1 The Utilisation Collapse

The near-halving of healthcare utilisation among non-members is the most striking finding and admits several explanations.

Cost barriers. Despite the expansion of primary healthcare services and the removal of user fees at public facilities for certain categories (pregnant women, children under 6), effective access may have declined due to transport costs, opportunity costs of queueing, and informal payments. The real decline in OOP spending among non-members could reflect price-rationed demand rather than improved coverage.

Public healthcare capacity constraints. The South African public health system has faced well-documented staffing shortages, infrastructure decay, and management challenges over the 2010–2023 period (Coovadia *et al.*, 2009; Maphumulo & Bhengu, 2019). If public facilities became less accessible or perceived as lower quality, non-members may have reduced their healthcare-seeking behaviour.

Survey methodology differences. The IES 2010/11 and 2022/23 differed in survey design, questionnaire structure, and data collection periods. The 2010/11 survey used diary-based recording of expenditure (weekly diaries for two weeks), which may capture more routine health purchases (e.g., over-the-counter medicines) than the recall-based approach of the 2022/23 survey. This methodological difference could account for part of the higher utilisation rates observed in 2010/11.

COVID-19 pandemic effects. The IES 2022/23 survey period (November 2022 to November 2023) followed the COVID-19 pandemic, which disrupted healthcare access patterns and may have persistently reduced healthcare-seeking behaviour, particularly among the uninsured.

4.2.2 The Gauteng Anomaly

Gauteng warrants specific discussion. It was the only province with a large, significant ATT in 2010/11 (R1,349 nominal, R2,330 in 2023 prices), which then *decreased* to R807 by 2022/23. This counter-trend may reflect Gauteng's unique position as the most urbanised and economically developed province, where private healthcare infrastructure was already extensive in 2010/11. The subsequent expansion of medical scheme membership to lower-income strata (facilitated by low-cost benefit options) may have brought in members with lower co-payment obligations, reducing the average ATT.

4.2.3 Changing Composition of Scheme Membership

The doubling of the proportion of households with medical scheme membership in the analysis sample — from 18.8% (21.8% weighted) in 2010/11 to 41.8% (42.5% weighted) in 2022/23 — is substantial. While part of this change may reflect the differing exclusion criteria and survey designs, it is consistent with the expansion of medical scheme products to lower-income segments. As the membership base broadens, the average socioeconomic profile of scheme members converges toward that of non-members (as evidenced by declining pre-matching SMDs), and the composition of co-payment obligations shifts.

The high weighted treatment rate of 42.5% in the 2022/23 sample — considerably above the population medical scheme coverage rate of approximately 16–17% (CMS, 2023) — likely reflects both the household-level measurement (where any member's scheme membership classifies the entire household as treated) and potential sample composition effects after exclusions.

4.3 Limitations

1. **Unobserved confounding:** NNM controls for observed covariates but cannot address unobserved confounders such as health status, chronic disease burden, risk preferences, or health literacy. Individuals in poor health may be both more likely to hold medical scheme coverage and more likely to incur health expenditure, biasing the ATT upward. This concern is more acute for the 2022/23 analysis where the ATT is significant.
2. **Limited common support in 2010/11:** The 66% match rate and residual SMD of 0.18 for log expenditure in 2010/11 indicate that the matched sample may not be representative of all treated households. The null ATT finding applies to the matchable subpopulation, which excludes the 34% of scheme members at the extreme upper tail of the expenditure distribution.
3. **Survey comparability:** The IES 2010/11 and 2022/23 differ in sampling frame (2001 Census vs GIF), survey period length, questionnaire design, and data collection methodology. The 2010/11 diary method may capture different types of health spending than the 2022/23 recall method, complicating direct comparisons. The different settlement type classifications (four categories in 2010/11 versus three in 2022/23) further limit strict comparability.
4. **CPI adjustment:** The inflation factor of 1.727 is based on headline CPI, which may not accurately reflect healthcare-specific price changes. Medical inflation in South Africa has

typically exceeded headline CPI, meaning that health-specific deflation would yield even larger real declines in OOP spending.

5. **Cross-sectional design:** Both surveys are cross-sectional, precluding panel analysis of the same households over time. The observed changes could reflect cohort effects, compositional shifts in the population, or genuine temporal trends.
6. **COICOP classification:** Some health-related expenditure may be misclassified (e.g., traditional medicine coded under Division 13), and the coding practices may have evolved between the two survey waves.

4.4 The Protection Paradox and SDG 3.8.2

The SDG 3.8.2 analysis reveals a fundamental limitation of financial risk protection indicators in contexts of demand suppression. The revised indicator — designed to be more pro-poor than the budget-share approach it replaces — still relies on *observed* OOP expenditure. When households forego care due to access barriers, their zero OOP is recorded as zero financial hardship. The indicator cannot distinguish between a household that spends nothing on healthcare because the public system provides free care (genuine protection) and one that spends nothing because it cannot access care at all (suppressed demand).

This creates the protection paradox identified in the results: the apparent improvement in national SDG 3.8.2 rates (from 30.6% to 14.8% under the WHO SPL) is driven substantially by the 47-percentage-point increase in zero-OOP households among the uninsured, not by actual improvements in financial protection. As Feedback 2 in the conceptual framework identifies, poverty suppresses demand, producing low OOP that the indicator misreads as protection.

The reversal — insured households now showing *higher* financial hardship than uninsured — is particularly consequential for policy. It suggests that as South Africa broadens insurance coverage through NHI, newly insured households may *increase* the national SDG 3.8.2 rate simply by accessing care they previously could not. This is a measurement paradox: successful expansion of coverage may register as *deterioration* in financial risk protection on the headline indicator.

4.5 Policy Implications

The findings have direct relevance for the National Health Insurance (NHI) policy debate:

- **The access gap is widening.** The dramatic decline in healthcare utilisation among non-members — from 88% to 41% of households with any health spending — suggests that the uninsured face intensifying barriers to care. The zero median OOP expenditure among matched non-members in 2022/23 is a stark indicator of unmet healthcare need.
- **Medical scheme co-payments represent a growing burden.** The emergence of a significant ATT by 2022/23 indicates that scheme membership now carries a meaningful OOP cost beyond premiums. Members face a double financial burden: premiums plus approximately R1,000 per year in co-payments.
- **Universal coverage design must address both gaps.** Any NHI system must simultaneously expand access for the currently uninsured (who are rationing care) and address the co-payment burden for scheme members (who are paying out of pocket despite insurance coverage).

- **The Gauteng convergence offers a positive signal.** The decline in Gauteng's ATT from R2,330 to R807 suggests that as scheme membership broadens and healthcare markets mature, the co-payment differential can moderate.
- **SDG 3.8.2 must be interpreted alongside utilisation data.** South Africa's apparently improving SDG 3.8.2 rate conceals a deepening access crisis. Policymakers should complement the headline indicator with utilisation-adjusted measures that distinguish suppressed demand from genuine financial protection. A household with zero OOP and no healthcare contact is not "protected" — it is excluded.

5 Conclusion

Using propensity score matching and Mahalanobis nearest-neighbour matching on two waves of the Income and Expenditure Survey, we document a structural transformation in the relationship between medical scheme membership and out-of-pocket health expenditure in South Africa. In 2010/11, both PSM (ATT = –R328, $p = 0.208$) and NNM (ATT = –R41, $p = 0.874$) find no significant co-payment differential. By 2022/23, PSM yields an ATT of R851 ($p < 10^{-37}$) and NNM yields R968 ($p < 10^{-43}$), both significant in all nine provinces. PSM achieves superior covariate balance (all post-matching SMDs < 0.02) and produces more conservative estimates than NNM, but the qualitative findings are identical. This emergence is driven not by increased scheme member spending but by a collapse in healthcare utilisation among non-members, whose utilisation rate fell from 95% to 46% among PSM-matched controls.

The OOP distribution among uninsured households is genuinely bimodal: 58.9% report zero health expenditure in 2022/23, up from 11.7% in 2010/11, with a clean gap between zero and the smallest positive values (R2). This confirms that the mass at zero represents true non-utilisation, not a rounding artefact.

Application of the revised SDG Indicator 3.8.2 (WHO, 2025b) reveals a protection paradox. Under the WHO Societal Poverty Line, the national financial hardship rate fell from 30.6% to 14.8% between the two surveys — but this apparent improvement is driven by demand suppression among the uninsured, not by genuine financial protection. A striking reversal has occurred: insured households now show higher financial hardship (17.4%) than uninsured households (12.7%), reflecting both the broadening of scheme membership to lower-income strata and the mechanical effect of zero OOP among non-utilisers. This reversal is robust across all four SPL variants tested and all nine provinces.

These findings underscore the urgency of addressing the widening healthcare access gap in South Africa and reveal a fundamental limitation of the revised SDG 3.8.2 indicator in contexts of demand suppression. Policymakers should complement the headline indicator with utilisation-adjusted measures. As South Africa implements NHI, the paradoxical implication is that successful coverage expansion may initially *increase* measured financial hardship by enabling care-seeking among previously excluded populations — a measurement challenge that must be anticipated in monitoring frameworks.

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